20 Martin did this question.

Rationalise the denominator of $\frac{14}{2+\sqrt{3}}$

Here is how he answered the question.

$$\frac{14}{2+\sqrt{3}} = \frac{14\times(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}$$

$$= \frac{28-14\sqrt{3}}{4+2\sqrt{3}-2\sqrt{3}+3}$$

$$= \frac{28-14\sqrt{3}}{7}$$

$$= 4-2\sqrt{3}$$

×	2	$-\sqrt{3}$
2	4	$-2\sqrt{3}$
$\sqrt{3}$	$2\sqrt{3}$	-3

Denominator = 4 − 3 = 1.

1 mark
The final product is negative

Martin's answer is wrong.

(a) Find Martin's mistake.

$\sqrt{3} \times (-\sqrt{3}) = -3$, not +3. The denominator is 1, not 7.

(1)

Sian did this question.

Rationalise the denominator of $\frac{5}{\sqrt{12}}$

Here is how she answered the question.

$$\frac{5}{\sqrt{12}} = \frac{5\sqrt{12}}{\sqrt{12}\times\sqrt{12}}$$

$$= \frac{5\times 3\sqrt{2}}{12}$$

$$= \frac{5\sqrt{2}}{4}$$

$\sqrt{12} = \sqrt{(4 \times 3)}$

$= 2\sqrt{3}$, not $3\sqrt{2}$.

Sian's answer is wrong.

(b) Find Sian's mistake.

She simplified $\sqrt{12}$ incorrectly: $\sqrt{12} = 2\sqrt{3}$, not $3\sqrt{2}$.

1 mark
Use the square factor 4

(1)

(Total for Question 20 is 2 marks)

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